

LexGuard — Full Foundations

Author: Jacob Iannotti — Date: September 2025

BLOCK 1 — Parts I–V: GP, DRO, Lyapunov+SoCPM, KKT/Margins, Domain Exemplars

PART I — GEOMETRIC PROGRAMMING (GP) DERIVATION FOR PARS-BUDGETED HARMONY

We maximize Harmony while budgeting (taxing or capping) risk via PARS and related factors. GP solves multiplicative structures globally after a log transform.

P1.1 Setup (Posynomial Form)

$H = (\text{Benefit} - \text{Burden}) - \text{Safety_Tax}$

$\text{Safety_Tax} = \alpha * \text{PARS} + \beta * \text{Gap} + \gamma * \text{Fragility}$

$\text{PARS} = \kappa * H_z * E_x * V_u * (1 - M_t)$ with $(H_z, E_x, V_u, 1 - M_t) > 0$

$\text{Benefit} = c_0 * Q / U$ with $Q = I * P$ (bilinear lift of UOE)

Approach A (Budgeted):

maximize Q / U

subject to $\kappa * H_z * E_x * V_u * (1 - M_t) \leq R$

$\text{Gap} \leq G$

$\text{Fragility} \leq F$

box bounds on $(I, P, U, H_z, E_x, V_u, M_t)$

Approach B (Penalized):

maximize $S = (Q/U) / [(\kappa * H_z * E_x * V_u * (1 - M_t))^\alpha * \text{Gap}^\beta * \text{Fragility}^\gamma]$

(We proceed with A.)

P1.2 GP-Eligibility

Standard GP: maximize monomial or minimize posynomial, with posynomial inequalities.

Objective Q/U is monomial if $Q = I * P$ and variables are positive.

PARS budget is a monomial $\leq \text{constant}$ (GP-compatible). Gap/Fragility upper bounds must be posynomial or bounded by posynomials.

Standing positivity: $I, P, U, H_z, E_x, V_u, (1 - M_t) > 0$.

P1.3 Log-Convex Reformulation

Let $y = \log x$ for positive x .

Monomial $c * \prod x_i^{a_i} \rightarrow \log c + \sum a_i y_i$ (affine)

Posynomial $\sum c_k * \prod x_i^{a_{ik}} \rightarrow \log \sum \exp(\log c_k + \sum a_{ik} y_i)$ (log-sum-exp convex)

Thus, maximize monomial $\Leftrightarrow$ minimize negative affine in y (convex). Constraints become convex after log.

P1.4 KKT System (Sketch)

Variables: $z = \log I$, $p = \log P$, $u = \log U$, $h = \log H_z$, $e = \log E_x$, $v = \log V_u$, $s = \log(1 - M_t)$.

$L = -(z + p - u) + \lambda_R * (\log \kappa + h + e + v + s - \log R)$

$+ \lambda_G * (\phi_G(y) - \log G) + \lambda_F * (\phi_F(y) - \log F)$

Stationarity equates marginal objective gains with weighted marginal risk increments; complementary slackness and feasibility complete KKT.

KKT multipliers become risk shadow prices; align with Harmony weights in penalized form.

PART II — DISTRIBUTIONALLY-ROBUST (DRO) DUALS: MOMENT & WASSERSTEIN

Goal: $\max_x \inf_{P \in \mathcal{P}_{\text{set}}} E_{\theta \sim P}[V(x; \theta)]$

D2.1 Moment-Based Ambiguity (Mean/Second Moment)

Theorem D2.1 (Linear-moment dual):

$\inf_{\{P: E[\phi] = m\}} E_P[V] = \sup_{\lambda} \{ \lambda^T m + \inf_{\theta} (V(x; \theta) - \lambda^T T) \}$

$\phi(\theta) \}$

Corollary (Bands): moment bands add L1 penalties on λ (support function of the band).

D2.2 Wasserstein Ambiguity

Theorem D2.2 (Kantorovich/Lipschitz bound): If V is L_c -Lipschitz under transport cost c ,

$$\inf_{\{P: W_c(P, \Phi) \leq \epsilon\}} E_P[V] = E_{\{\Phi\}}[V] - L_c \epsilon$$

General dual: $\sup_{\{\lambda \geq 0\}} \{ E_{\{\Phi\}}[\inf_{\theta} (V(\theta) + \lambda c(\theta, \tilde{\theta}))] - \lambda \epsilon \}$

PART III — LYAPUNOV-STYLE CERTIFICATES COUPLING V^* DYNAMICS & SoCPM

Dynamics: $V_{t+1} = ((I_t P_t - (W_t + \epsilon))/U_t) E + F_t$ with $|F_t| \leq F_U, 0 \leq E \leq 1$

Safe set $S = \{ x: H(x) \geq 0, \min_i M_{\{g_i\}}(x) \geq \delta \}$

SoCPM Redirect applies policy $K(x_t)$ when $x_t \notin S$.

L3.2 Candidate Lyapunov and Negative Drift

$$L(x) = a[H(x)]_- + b[\delta - \min_i M_{\{g_i\}}(x)]_+ + c \text{PARS}(x)$$

Assume Redirect improves H , margins, and shrinks PARS by factor $\rho < 1$, and F_t bounded.

Theorem L3.1: $\exists a, b, c > 0$ s.t. $L(x_{t+1}) - L(x_t) \leq -\epsilon_L < 0$ whenever $x_t \notin S$. $\Rightarrow$ Safety invariance.

PART IV — KKT/DUAL INTERPRETATION OF MARGIN MAPS AS SHADOW PRICES

Problem: maximize $H(x)$ s.t. $g_i(x) \geq 0$.

KKT: $\nabla H = \sum_i \lambda_i \nabla g_i, \lambda_i \geq 0, \lambda_i g_i = 0$.

Margin $M_{\{g_i\}}(x) = g_i(x) / \|\nabla g_i(x)\|$.

Theorem K4.1: marginal Harmony gain per unit increase in i -th margin equals λ_i (shadow price).

PART V — DOMAIN EXEMPLARS (NUMERIC, END-TO-END)

V5.1 Legal/Compliance: $I=4, P=0.7, W=1, U=2 \Rightarrow V=0.9; \text{PARS}=0.3; \text{Tax}=0.54; \text{Burden}=0.1 \Rightarrow H=0.26$.

SoCPM thresholding shows tone/mitigation trade for Proceed/Redirect.

V5.2 Healthcare: $V=0.6$; initial $H < 0$; increase $M_t \rightarrow 0.8 \Rightarrow H > 0$; clears SoCPM.

V5.3 Climate: $V=0.333$; initial $H < 0$; $M_t 0.3 \rightarrow 0.8 \Rightarrow H > 0$; SoCPM Proceed.

BLOCK 2 — Five Additional Math Sections + Consolidation

A1) Multi-PARS Portfolio via GP with Couplings

$$\text{PARS}_k = \kappa_k \text{Hz}_k \text{Ex}_k \text{Vu}_k (1 - M_t)_k$$

$$S_k = (Q_k / U_k) * \prod_j \text{Penalty}_{\{k,j\}}^{-w_{\{k,j\}}}, Q_k = I_k * P_k$$

maximize $\prod_k S_k^{\omega_k}$

subject to $\sum_k \text{PARS}_k \leq R_{\text{tot}}, \sum_k \text{Gap}_k \leq G_{\text{tot}}, \sum_k \text{Fragility}_k \leq F_{\text{tot}},$

and couplings: $\sum_k c_k (1 - M_t)_k \leq B, \sum_k I_k \leq I_{\text{tot}}.$

After log: affine objective + log-sum-exp constraints $\Rightarrow$ convex. KKT multipliers are portfolio shadow prices.

A2) Wasserstein-2 DRO with Lipschitz Moduli

Setup: $\max_x \inf_{\{P: W_2(P, \Phi) \leq \epsilon\}} E_P[V(x; \theta)],$ with cost $\|\theta - \tilde{\theta}\|_2^2$.

Theorem A2.1: If V is L -Lipschitz in θ , $\inf_{\{W_2 \leq \epsilon\}} E[V] \geq E_{\{\Phi\}}[V] - L \epsilon$.

Dual form: $\sup_{\{\lambda \geq 0\}} \{ E_{\{\Phi\}}[\inf_{\theta} (V + \lambda \|\theta - \tilde{\theta}\|_2^2)] - \lambda \epsilon^2 \}.$

A3) Composite Lyapunov with Tone (Ar) & Exposure (Ex) for De-escalation

$$L(x) = a[H]_- + b[\delta - \min_i M_{\{g_i\}}]_+ + c \text{PARS} + d \psi(\text{Ar}) + e \chi(\text{Ex})$$

Redirect ensures: $H \uparrow$ by η_H , margins $\uparrow$ by η_M , $\text{PARS}^*(t+1) \leq \rho \text{PARS}_t$, $\text{Ar}_{t+1} \leq$

$\rho_{Ar} \cdot Ar_t, Ex_{\{t+1\}} \leq \rho_{Ex} \cdot Ex_t$.

Theorem A3.1: $\exists \text{ weights} \Rightarrow \Delta L \leq -\epsilon_c < 0$ outside $S \Rightarrow$ formal de-escalation.

A4) Second-Order Sensitivity (Parametric LP Bands / Smooth Hessians)

Within a stable LP basis B^* : $J(\theta) = J(\theta_0) + \sum_i \pi_i^* \Delta \theta_i$.

Band: $J \in [J_0 - \sum \pi_i^* \|\Delta \theta_i\|, J_0 + \sum \pi_i^* \|\Delta \theta_i\|]$.

Basis flips capture curvature; smooth surrogate (softplus on McCormick) yields Hessians if needed.

A5) Multi-Period Worked Case with Uncertainty Sets (Auditor-Ready)

For $t=1..T$: $V_t = (I_t \cdot P_t - W_t) / U_t$; $PARS_t = \kappa \cdot Hz_t \cdot Ex_t \cdot Vu_t \cdot (1 - Mt_t)$;

$H_t = (V_t - Burden_t) - [\alpha \cdot PARS_t + \beta \cdot Gap_t + \gamma \cdot Fragility_t]$.

Objective: $J = \inf_{\{\theta_t \text{ in } U_t\}} \sum \delta^{t-1} H_t(x_t; \theta_t)$, with SoCPM policy.

Certificate pack: LP/GP logs, DRO certs, Lyapunov drift, margin series, Redirect traces.

B) Consolidation Snapshot

Covered (12 sections): UOE; Harmony; PARS; SoCPM; ILC; Gap/Tethered; Margin Maps; Portfolio GP; DRO; Composite Lyapunov; 2nd-Order Sensitivity; Multi-Period Case.

Missing (now addressed in Block 3): Provenance/Lineage; SBOM Gate; Traceability; Adversarial calculus; Calibration; Complexity; Implementation correctness; Legal alignment.

BLOCK 3 — Sections 13–20 Fully Written (Governance & Ops)

13) Provenance & Lineage Ledger — Formal Model and Guarantees

Objects and commits:

Capsule $C = (\text{DataVer}, \text{ModelVer}, \text{GuardrailVer}, \text{Seed}, \text{Config}, \text{Runtime})$

$\text{Commit}(a) = H(\text{bytes}(a) \parallel C \parallel \text{Parents}(a))$

Ledger: DAG of commits with Merkle forest for $O(\log N)$ inclusion proofs.

Theorem 13.1 (Minimal blame landing): Build auxiliary s-t flow on lineage DAG; a minimum cut intersects all root $\rightarrow$ incident paths $\Rightarrow$ minimal landing set; polynomial time.

Theorem 13.2 (Binding): Collision-resistant hash $\Rightarrow$ cannot alter a or C without changing $\text{Commit}(a)$.

14) SBOM Gate for AI — Release Policy as Mathematics

SBOM $S(r) = (\text{ModelVer}, \text{DataSlices}, \text{GuardrailSet}, \text{EvalSuite}, \text{RiskBudget})$.

Gate rules: signatures; $\sum PARS(a) \leq R_{tot}$; eval metrics $\geq$ thresholds τ_m ; calibrated T for SoCPM.

Theorem 14.1 (Ex-ante/ex-post soundness): If gate holds and SoCPM uses calibrated T , shipped artifacts either satisfy $H \geq 0$ on validation or are redirected at runtime.

Theorem 14.2 (Calibration ROC band): $\exists [T_-, T_+]$ yielding monotone trade between false-blocks and false-passes.

15) Requirements Traceability — Spec $\rightarrow$ Proofs $\rightarrow$ Tests $\rightarrow$ Metrics

Trace matrix R maps each requirement to: theorems (T), tests (X), metrics (M) with ledger pointers.

Theorem 15.1 (Audit completeness): If every requirement has linked theorem + passing tests + metrics within tolerances, the system is audit-complete.

16) Adversarial Model & Red-Team Calculus

Threat sets $\Delta_{prompt}, \Delta_{eval}, \Delta_{data}, \Delta_{jail}$ (compact). Robust objective uses tethered value:

$\tilde{V}(x) = \inf_{\{\delta \in \Delta\}} V(x; \delta)$; $\tilde{H}(x) = (\tilde{V} - Burden) - \text{Safety_Tax}$.

Theorem 16.1 (Existence): Under compactness/continuity on compact feasible set, $\max_x \tilde{V}$ and $\max_x \tilde{H}$ exist.

Theorem 16.2 (Coverage lower bound): With coverage q and Lipschitz V , $\inf_{\Delta} V \geq \inf_S V - L \cdot (1-q) \cdot \text{diam}(\Delta)$.

17) Calibration Protocol — (α, β, γ, T) from Data

Fit weights via NNLS: $E[L] \approx \alpha \cdot E[PARS] + \beta \cdot E[Gap] + \gamma \cdot E[Fragility]$.

Theorem 17.1: NNLS preserves monotonicity $\Rightarrow$ Harmony nonincreasing in risk components.

Threshold T chosen by validation ROC; PAC-like bound with VC-dim d and n samples:

$\text{TrueError}(T) \leq \text{EmpError}(T) + O(\sqrt{(d \cdot \log n + \log(1/\delta)) / n})$.

18) Complexity & Runtime — Scaling Laws

Lifted UOE LP: polynomial time (interior-point or simplex).

Portfolio GP: convex, $\sim O(p^3)$ with sparsity benefits.

DRO duals: small multiplier dimension overhead.

Multi-period: $\sim$ linear in T when separable; coupling adds convex constraints.

19) Implementation Correctness — Capsules, Numerics, Tolerances

Capsule: pinned container, code/weights/data digests, seeds, solver versions/tolerances.

Theorem 19.1: Reproducibility up to epsilon for Lipschitz functionals with fixed tolerances.

Guard against silent violations: interval arithmetic on guardrails; accept only if lower bound ≥ 0 .

Theorem 19.2: Interval evaluation prevents false passes due to rounding.

20) Legal Alignment Appendix — ALARP & Duty of Care

ALARP mapping: maximize $H = (\text{Benefit} - \text{Burden}) - \alpha * \text{PARS} - \beta * \text{Gap} - \gamma * \text{Fragility}$
implies mitigation increases until priced marginal risk reduction equals marginal burden.

Duties: Runtime (SoCPM), Ex-ante (SBOM Gate), Ex-post (Lineage/Blame landing). Together $\Rightarrow$ measurable, auditable standard of care.